

- 1 Figure 1.1 is a transmission electron micrograph of a section through part of the pancreas of a mammal.

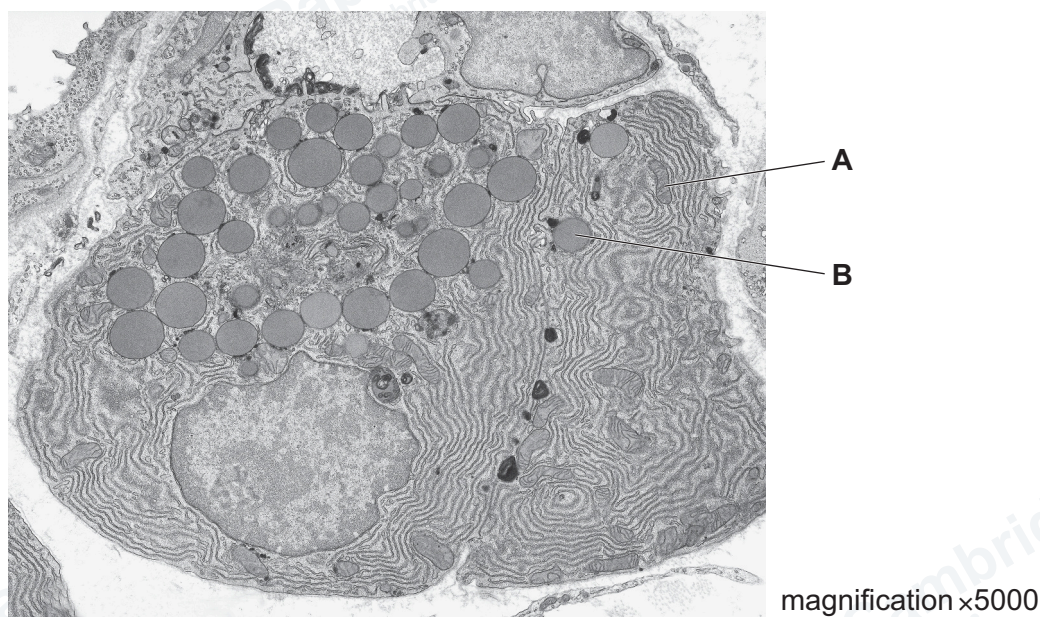


Figure 1.1

- (a) (i) Describe the steps to determine the actual length in micrometres (μm) of the organelle labelled **A** in Figure 1.1.

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..... [2]

- (ii) The structures labelled **A** and **B** are visible when the cells in Figure 1.1 are viewed using a light microscope.

State **two other** structures in the cells shown in Figure 1.1 that are visible when using a light microscope.

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..... [1]



- (iii) Explain why some of the cell structures that are visible with the transmission electron microscope are **not** visible when using the light microscope.

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..... [2]

- (b) The main function of the cells shown in Figure 1.1 is to synthesise and release enzymes that digest proteins, lipids and carbohydrates in the digestive system.

State the cell structures visible in Figure 1.1 that are involved with the synthesis and release of enzymes **and** outline the role of each structure in these processes.

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..... [3]

[Total: 8]